Why WavLM (vs HuBERT) for the CAARMA critic?

Both: 24-layer Transformer encoder · hidden 1024 · 315M params

HuBERT (Hsu et al. 2021)	**WavLM (Chen et al. 2022)**
Pretrain corpus: LibriLight 60kh (English read speech)	Pretrain corpus: 94kh (LibriLight + GigaSpeech + VoxPopuli)
Pretraining objective: Masked prediction of k-means cluster IDs (focus = phoneme-level acoustic content)	**Pretraining objective: Masked prediction + utterance MIXING + denoising (designed for speaker tasks)**
What hidden states encode: low layers : acoustic features mid layers : phonemes high layers : phoneme + some speaker info	What hidden states encode: low layers : acoustic + noise robustness mid layers : phonemes + speaker identity high layers : strong speaker discriminative
CAARMA layer choice {h7, h9, h11, h12} → reaches for the SPEAKER-rich mid/high layers	**Same layer choice {h7, h9, h11, h12} → but layers are inherently MORE speaker-discriminative**
SUPERB SID: 81.4% ASV EER: 5.98%	**SUPERB SID: 95.5% ASV EER: 3.77%**

Empirical finding (CAARMA, this work): WavLM 3.71% vs HuBERT 3.48% EER
→ The seq_len = 1 input collapses WavLM's sequence-level advantages.